

Price Forecasting

Based on my experience

There Is No Singular Perfect Model

A. There is no singular right answer.

- There are infinite number of ways for forecasting price.
- Getting feedback across multiple groups within the organization, and building a consensus on how to approach this is crucial.

B. But, there could be some wrong answers.

- Multi Collinearity Problem: Higher correlation between x variables should be avoided.
(E.g., it might not be a good idea to include NASDAQ, S&P500 index at the same time)
- Omitted Variable Bias: Omitting important indicators can result in unreliable models.
(E.g., relying solely on on-chain data may overlook the broader economic factors that can influence the coin's price)

Model Should Evolve

A. Influencing factors change as time goes on.

- For instance, I've seen Bitcoin Spot ETF launch in Jan 2024, and the total balance of them is highly correlated with Bitcoin price.
- Transaction counts become less influential to the Bitcoin price.

B. Model should be updated and shared with relevant stakeholders regularly.

- Updating the model every n^{th} Friday of month
- "What explains the difference between the previous forecast and the updated forecast?"
- "What factors are newly added or excluded in this iteration, and why?"

(Almost) All Possible Indicators Should Be Considered

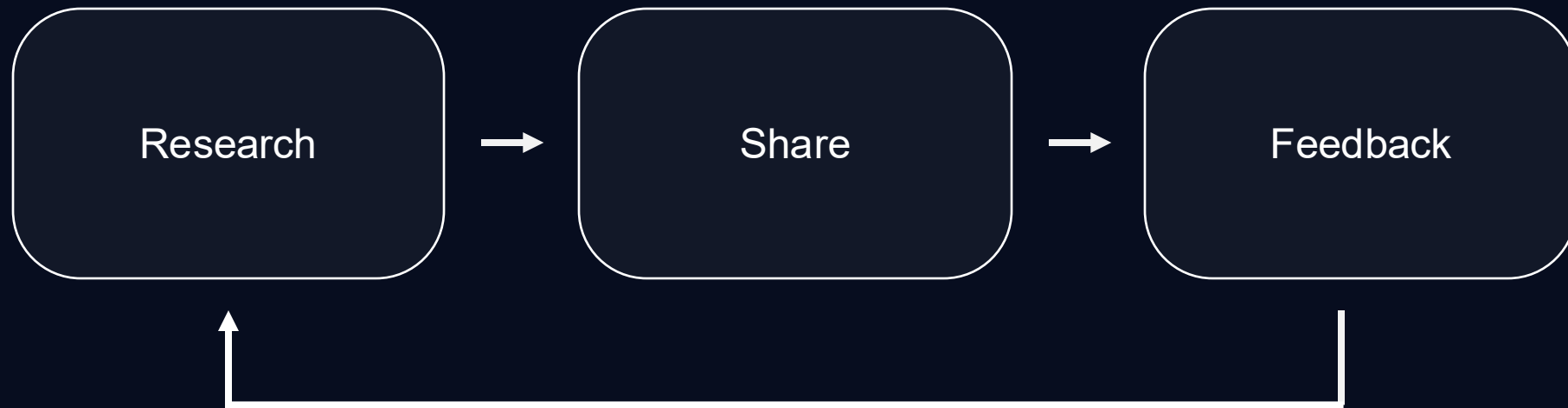
A. Potential Indicators for Coin Price

- Broader Crypto Market: Top 100 cryptocurrency price & supply, BTC price, ETH price
- Onchain Data: # of active address, # of github stars, # of github pulls, # of twitter mentions
- Macro Economic Factor: Interest rates, inflation, savings rate, employment, payroll, etc.
- Current Market Condition: Current coin price, historical price (30MA, 60MA, 90MA, etc)

B. Technical Advice

- Target p-value at least ≤ 0.2
- Exclude any highly correlated x variables (≥ 0.8)
- In the end, 5-8 indicators (no more than 10) work in most cases.
- Select data ranges well; it's usually sensitive to what timeframe is used for modeling.

Research, Share, Feedback & Repeat



This should be a **collaborative effort** with other groups. If the model's forecasts are off from the actuals, the message should be **"Let's see what elements of our model are not functioning well. Let's update it together!"**